Retail Sales Forecasting

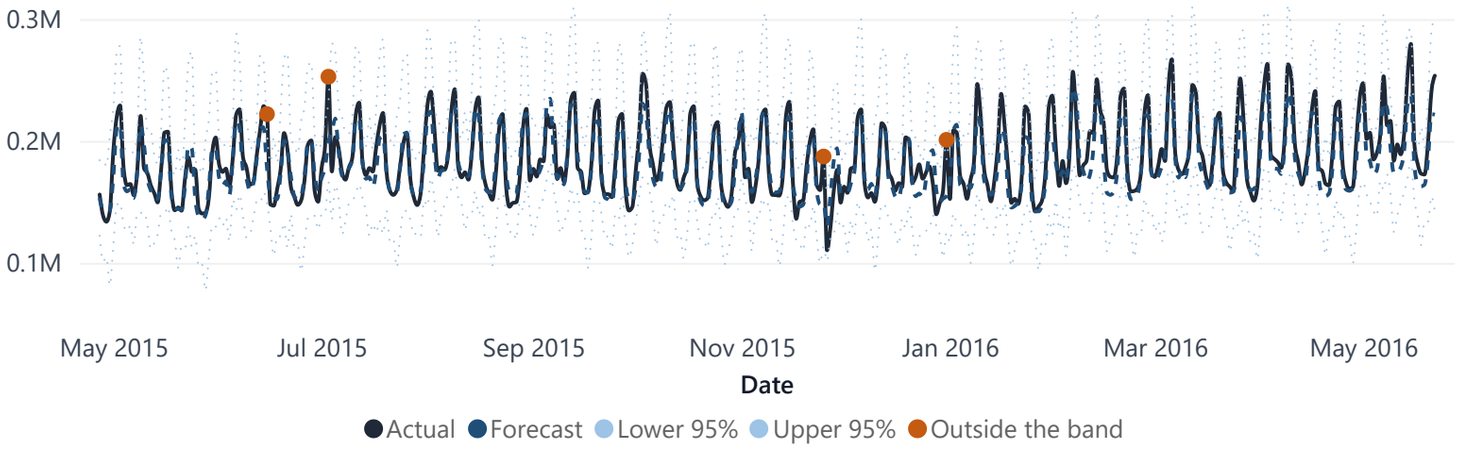
Hybrid ARIMA-LSTM · test period · scored days only · store managers: next 28 days for one store

Forecast | Models | Planning

Dataset: M5 | Store: All | Model: All | Forecast origin (number): All

All stores · All models · select one origin

Actual (solid) and forecast (dashed) daily sales with the 95% interval (dotted); orange dots = outside the band



MASE

0.884

Seasonal naive 1.196

INTERVAL COVERAGE

92.8%

Target 95%
(green within ±2 points)

MAE

323.2

units sold per day

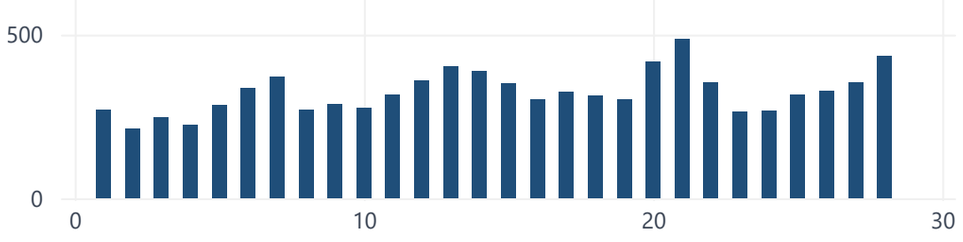
BIAS

-2.7%

Positive = over-forecast

MAE by days ahead

Error grows with the horizon (metric follows the Models page switch; default MAE)



Daily detail

Sorted by date · orange = actual outside the 95% interval

Date	Horizon	Actual	Forecast	Lower	Upper	Inside (1 = yes)
2015-04-27	1	156,072	153,207	122,225	184,188	1
2015-04-28	2	138,833	143,075	101,119	185,032	1
2015-04-29	3	133,779	142,094	104,716	179,471	1
2015-04-30	4	142,581	143,001	81,515	204,505	1

"Accuracy on scored days · 95% split-conformal intervals · Units: M5 = units sold, Rossmann = turnover · Source: dashboard/data export"

Retail Sales Forecasting

Hybrid ARIMA-LSTM · test period · scored days only · planners: which model, at which horizon, for which store

Forecast | Models | Planning

Dataset: M5 | Store: All | Model: All | Forecast origin (number): All

BEST MODEL (MASE)

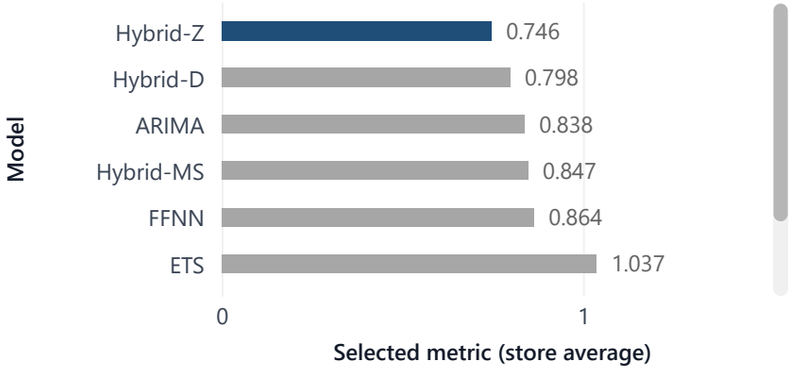
Hybrid-Z

MASE 0.746 · lowest in the selection

Metric: MASE | RMSE | MAE | MAPE

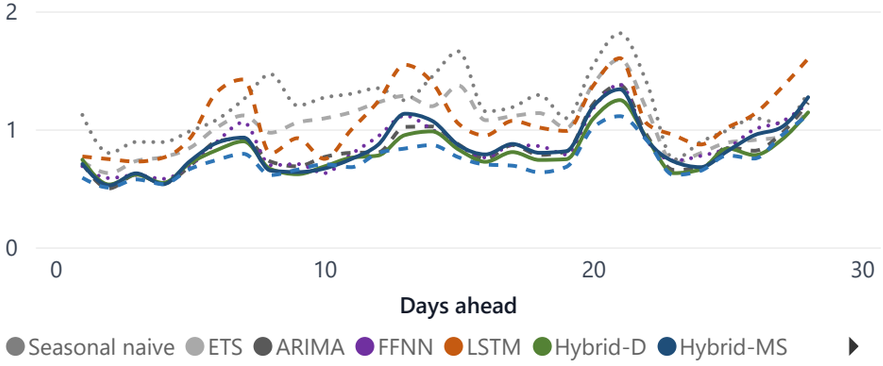
Store-average MASE by model · lower is better

Best model in dark blue, others grey (store average, all test origins)



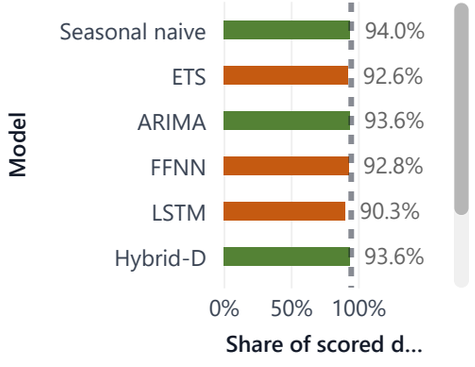
MASE by days ahead

One line per model; line style and colour both differ



Interval coverage by model

Share of days inside the 95% interval · li...



MASE by store and model

Darker = higher error · store average over the test origins

Store	Seasonal naive	ETS	ARIMA	FFNN	LSTM	Hybrid-D	Hybrid-MS	Hybrid-Z
CA_1	0.980	0.830	0.682	0.715	0.703	0.655	0.678	0.572
CA_2	1.653	1.666	1.648	1.735	2.991	1.533	1.797	1.439
CA_3	0.987	0.907	0.649	0.650	0.693	0.626	0.630	0.562
CA_4	1.093	0.898	0.851	0.975	1.093	0.827	0.894	0.780
TX_1	1.041	0.865	0.619	0.669	0.677	0.612	0.646	0.579
TX_2	1.135	0.865	0.711	0.699	0.688	0.687	0.702	0.649
TX_3	1.272	1.062	0.770	0.870	0.988	0.747	0.804	0.708

"Accuracy on scored days · 95% split-conformal intervals · Units: M5 = units sold, Rossmann = turnover · Source: dashboard/data export"

Retail Sales Forecasting

Hybrid ARIMA-LSTM · test period · scored days only · planners: stock decisions from the forecast range

Forecast | Models | Planning

Dataset: M5 | Store: All | Model: All | Forecast origin (number): All

FORECAST SALES

70,733,426

units sold

SAFETY STOCK PROXY

17,911,476

25.3% of forecast sales

STOCK-OUT RISK DAYS

1,044.7

Actual above the upper bound

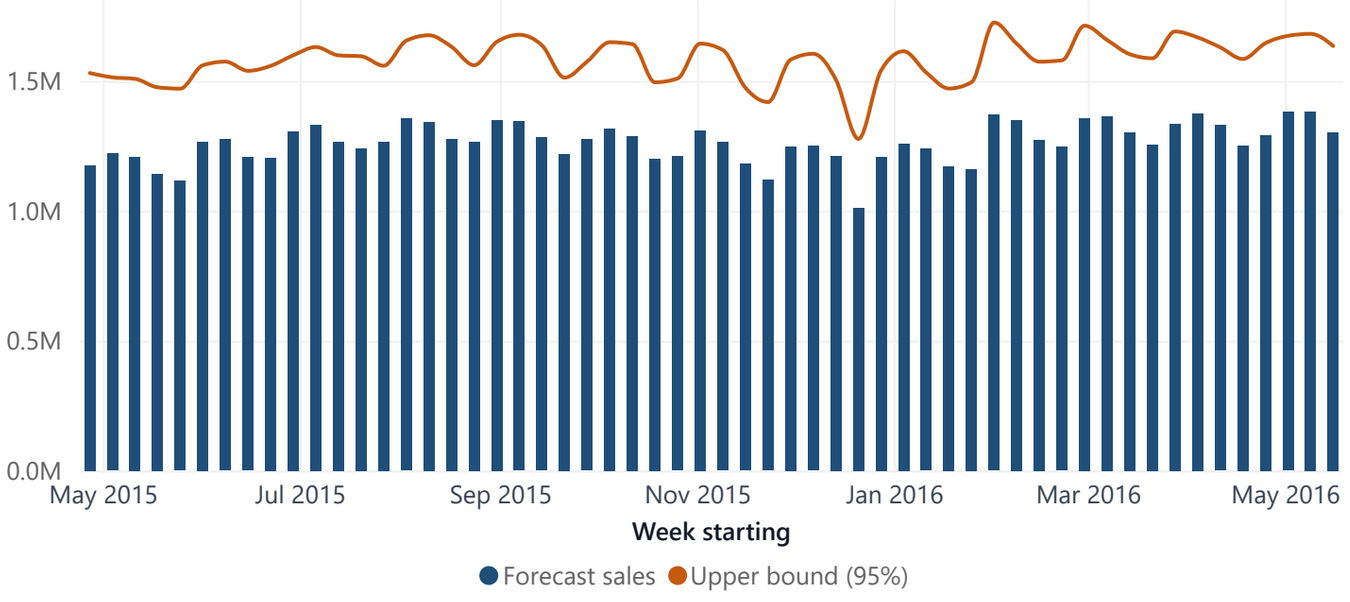
OVERSTOCK RISK DAYS

261.7

Actual below the lower bound

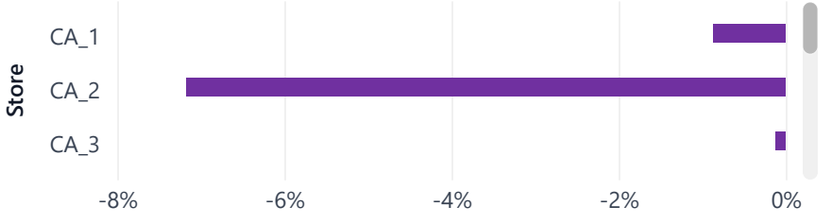
Weekly forecast and upper bound

Plan to the line, not the bar · drill down from week to day (arrows in the header)



Bias by store

Under-forecast (purple) ◀ 0 ▶ over-forecast (orange)



Stores by stock-out risk

Sorted by stock-out risk days

Store	Stock-out days	Overstock days	Safety stock
CA_2	446	62.5	1,066,367
WI_1	156.3	54.5	1,290,048
CA_4	102.2	31	803,295
Total	**1,044.7**	**261.7**	**17,911,476**

"Accuracy on scored days · 95% split-conformal intervals · Units: M5 = units sold, Rossmann = turnover · Source: dashboard/data export"